

HARDWARE RETAIL × DATABRICKS · OCTOBER 09, 2026

Improving product retrieval on Databricks AI Search

600 labelled shopper queries · 38,291 SKUs · 48 retrieval strategies, all traced and scored in MLflow.

0.661

baseline MRR@10 (HYBRID)

0.939

best MRR@10 (17_ltr_cv)

+0.279

absolute gain

The ask

- Improve retrieval accuracy on **structured retail product data** where shoppers mix intent, brand, specs and exclusions.
- Show which Databricks AI Search capabilities move the needle, with **measured** evidence rather than anecdotes.
- Give a reproducible, governed evaluation harness the retailer can run on its own catalog.

How we measured

Data & index

- `retail_consumer_goods.product_search.products` : 38,291 products, 500+ merchandise classes
- Delta Sync AI Search index, managed `databricks-gte-large-en` embeddings, ANN / FULL_TEXT / HYBRID
- MLflow evaluation dataset
`retail_consumer_goods.product_search.retrieval_eval` : 600 queries, 6 types, ground truth from the catalog

Metrics (MLflow scorers)

- **hit@k, MRR@10, nDCG@10**: is the right product found, and how high
- **constraint precision@10**: share of results honoring brand / class / exclusion
- **excluded-brand leak**: “not DeWalt” but DeWalt shown
- **latency** (p50 / p90 / p99 from traces), LLM & ai_decide call counts
- **Correctness** + retrieval judges on a 100-query subset

Methods compared

family	what it does	Databricks capability
Plain RAG	query → top-10	<code>VectorSearchRetrieverTool</code> ANN / FULL_TEXT / HYBRID
Reranker	cross-encoder reorders candidates; columns fixed or targeted to the request	<code>DatabricksReranker(columns_to_rerank=...)</code> + <code>ai_decide</code> column profile
Dynamic filtering	LLM turns brand/exclusion/class intent into index filters	<code>VectorSearchRetrieverTool(dynamic_filter=True)</code> + gpt-oss-120b
Instructed retrieval	every candidate judged against the request + store policy	<code>ai_decide</code> REST (score / noul questions)
dao-ai instructed	LLM decomposition into filtered subqueries → RRF → rerank	<code>dao_ai.tools.create_ai_search_tool</code> + <code>dao_ai.config</code>
Router (09)	SKU/UPC → exact lookup; "not X" → brand NOT filter; else cleaned HYBRID; optional <code>ai_decide</code>	<code>VectorSearchRetrieverTool</code> filters + <code>ai_decide</code> , no generative LLM
Enriched index (10)	embed a compact title brand class attributes codes field	second Delta Sync index on <code>search_text</code>
Cross-encoder rerank (13)	open rerankers score router candidates on GPU	bge-reranker-v2-m3 / Qwen3-Reranker on Model Serving
Doc2Query (15)	LLM-written shopper searches indexed with each product	offline generation + new index / RRF fusion
Query plan (16)	LLM parses hard vs soft constraints, cached	gpt-oss-120b tool call, catalog-validated
Learned ranker (17)	LambdaRank over retrieval, <code>ai_decide</code> , cross-encoder and lexical signals	LightGBM, 5-fold CV + holdout
Facet-guided	facet counts give the real filter vocabulary; <code>ai_decide</code> picks	AI Search <code>facets</code> + <code>ai_decide</code> choice

State-of-the-art patterns tested

All models are Databricks-hosted or served on Databricks; all vectors live in Databricks AI Search.

pattern	industry evidence	implementation	notebook
Instruction-aware embeddings	Qwen3-Embedding (MTEB top tier)	databricks-qwen3-embedding-0-6b, self-managed vectors	14
Open cross-encoder rerankers	ShopRank-Bench 2026; Qwen3-Reranker, bge-reranker-v2-m3	Model Serving GPU endpoints	12-13
Doc2Query expansion	AliExpress SAM-D2Q (+3.4% GMV); Doc2Query++ dual-index fusion	offline LLM generation + RRF	15
Structured query plans	Instacart Intent Engine; Amazon hint-augmented reranking	gpt-oss-120b plan, cached, validated	16
Listwise reranking	RankGPT (EMNLP 2023)	ai_decide choice over top 5	17
Learned ranking / distillation	Walmart LLM-labelled dense retrieval (+5.1% nDCG); Etsy <10 ms student	LightGBM LambdaRank	17
Graded relevance evaluation	Amazon ESCI; bias-corrected LLM judging	Claude Sonnet ESCI judge + holdout set	11, 19
Billion-scale vector search	Databricks storage-optimized endpoints (~1B vectors)	5M-row storage-optimized test	18

Why reranking alone moved the average less than expected

1. **Identifier lookups are invisible to relevance ranking.** 100 of 600 queries are SKU/UPC lookups; every strategy that only ranks text scored at most MRR 0.00 on them, which caps the overall average.
2. **HYBRID leaves little headroom on the rest.** Of 500 non-identifier queries the target is already at rank 1 for 350; only 129 have it at rank 2–10, the only slice a reranker can improve.
3. **Rerankers can only reorder.** When the excluded brand fills the candidate pool, reranking cannot remove it; only a filter can.
4. **Coarse decisions tie.** A 4-level `score` question leaves several candidates tied at the top, so order falls back to retrieval; continuous `norml` probabilities separate them.
5. **A cross-encoder is not instruction-following.** It matches the words in the request, including the brand being excluded, and treats pack-size variants as equivalent.

Leaderboard (all 600 queries)

strategy	hit@1	hit@10	MRR@10	nDCG@10	constraint prec@10	excluded-brand leak	zero results	p50_ms	p90_ms	p99_ms
17_ltr_cv	90.2%	99.2%	0.939	0.950	76.7%	1.0%	0.0%	0.000	0.000	2.000
13_router_bge	89.3%	99.0%	0.930	0.943	76.7%	1.0%	0.0%	596	1,149	3,433
15_d2q_router_quality	88.7%	99.2%	0.928	0.941	77.1%	1.0%	0.0%	1,230	1,913	4,258
16_plan_router_quality	88.7%	98.8%	0.925	0.939	73.0%	12.0%	0.0%	1,790	2,477	4,327
20_ai_router_bge	88.5%	98.3%	0.923	0.936	74.7%	10.0%	0.0%	802	1,742	9,564
17_router_listwise	87.8%	99.2%	0.919	0.935	76.6%	1.0%	0.0%	1,010	1,641	3,670
14_qwen3_router_quality	87.2%	98.8%	0.914	0.932	77.0%	1.0%	0.0%	1,260	1,714	4,248
10_enriched_router_quality	86.5%	98.5%	0.911	0.928	76.7%	1.0%	0.0%	1,243	2,508	14,581
13_router_qwen3	84.5%	99.0%	0.902	0.922	77.1%	1.0%	0.0%	5,419	7,568	9,096
20_ai_router_quality	84.5%	97.7%	0.897	0.913	74.2%	12.0%	0.0%	1,051	1,967	5,284
09_router_quality	85.2%	98.2%	0.896	0.915	76.3%	1.0%	0.0%	1,225	1,525	3,020
10_enriched_router_fast	82.7%	98.5%	0.885	0.908	76.5%	1.0%	0.0%	266	355	1,170
16_plan_router_fast	82.7%	98.7%	0.884	0.908	72.9%	12.0%	0.0%	295	966	3,312
07_dao_ai_instructed_llm_rerank	82.5%	94.3%	0.869	0.885	85.6%	2.0%	0.0%	7,923	13,828	19,490
09_router_fast	78.8%	98.0%	0.861	0.886	76.0%	1.0%	0.0%	261	512	2,691
20_ai_router_fast	77.8%	97.7%	0.852	0.879	74.5%	12.0%	0.0%	640	2,398	12,455
09_hybrid_id_routing	75.0%	96.5%	0.827	0.857	66.7%	79.0%	0.0%	284	2,028	4,859
18_scale_router_bge_50	75.5%	84.7%	0.783	0.789	93.2%	1.0%	0.0%	1,729	2,304	3,491
13_hybrid50_bge	71.0%	81.5%	0.750	0.764	68.0%	80.0%	0.0%	2,554	2,716	3,765
07_dao_ai_instructed_flashrank	68.8%	81.5%	0.738	0.749	78.1%	1.0%	0.0%	17,626	28,667	38,090
06_filter_ai_decide	65.2%	87.2%	0.734	0.764	72.5%	1.0%	0.0%	1,567	2,473	5,651
10_enriched_rerank	64.8%	86.7%	0.726	0.760	63.0%	93.0%	0.0%	465	599	1,190
10_enriched_full_text	64.3%	88.0%	0.726	0.761	66.1%	80.0%	0.0%	108	143	264
06_ai_decide_noul	67.0%	80.3%	0.721	0.738	68.9%	66.0%	0.0%	4,482	36,440	70,971
09_ai_decide_fast	66.7%	79.8%	0.713	0.732	66.7%	79.0%	0.0%	1,262	1,950	4,106
18_scale_router_quality_50	68.0%	79.7%	0.713	0.722	91.7%	1.0%	0.0%	18,510	73,765	142,269

strategy	hit@1	hit@10	MRR@10	nDCG@10	constraint prec@10	excluded-brand leak	zero results	p50_ms	p90_ms	p99_ms
13_hybrid50_qwen3	64.5%	80.5%	0.707	0.730	65.2%	86.0%	0.0%	11,338	12,504	37,389
15_d2q_hybrid	64.0%	81.2%	0.703	0.727	67.7%	80.0%	0.0%	285	420	1,375
15_fusion_enriched_pseudoq	63.8%	81.3%	0.702	0.728	67.1%	82.0%	0.0%	332	702	1,816
10_enriched_hybrid	62.3%	82.3%	0.689	0.719	67.2%	80.0%	0.0%	262	506	2,586
05_dynamic_filter_guarded_rerank	58.3%	88.3%	0.680	0.728	73.6%	1.0%	0.0%	1,635	3,622	6,435
03_full_text	60.7%	80.0%	0.677	0.705	65.7%	79.0%	0.0%	127	167	343
05_dynamic_filter_guarded	58.0%	88.0%	0.676	0.721	73.3%	1.0%	0.0%	1,118	3,413	6,237
14_qwen3_hybrid	59.0%	84.7%	0.671	0.713	64.1%	86.0%	0.0%	293	447	621
18_scale_router_quality	63.8%	73.5%	0.667	0.674	93.5%	1.0%	0.0%	1,223	2,096	4,331
06_ai_decide_score	60.0%	78.2%	0.664	0.686	66.1%	74.0%	0.0%	4,231	50,000	112,433
03_hybrid	58.3%	79.8%	0.661	0.690	66.7%	79.0%	0.0%	291	1,310	6,417
14_qwen3_hybrid_noinstr	57.0%	83.3%	0.651	0.694	64.4%	87.0%	0.0%	300	466	658
07_dao_ai_instructed	55.2%	81.8%	0.648	0.688	78.3%	1.0%	0.0%	1,701	3,367	4,966
05_dynamic_filter	54.5%	84.5%	0.645	0.690	70.4%	1.0%	3.7%	957	1,785	4,522
04_rerank_name_desc	55.0%	80.5%	0.642	0.680	65.9%	87.0%	0.0%	1,174	4,252	8,513
04_rerank_targeted	53.0%	77.0%	0.612	0.649	66.1%	86.0%	0.0%	814	1,985	8,225
03_ann	51.8%	76.7%	0.606	0.640	65.3%	71.0%	0.0%	253	876	5,467
18_scale_router_fast	55.3%	72.5%	0.598	0.619	93.5%	1.0%	0.0%	716	1,111	2,200
04_rerank_all	49.0%	78.3%	0.590	0.633	63.6%	91.0%	0.0%	528	791	3,398
14_qwen3_ann	49.8%	72.8%	0.572	0.611	60.0%	86.0%	0.0%	284	461	659
08_facet_guided	47.3%	63.8%	0.535	0.558	70.6%	9.0%	0.0%	2,516	5,155	9,269
18_scale_hybrid	32.2%	53.7%	0.380	0.409	67.9%	97.0%	0.0%	818	2,165	4,179

Are the gains real?

Paired bootstrap over the same queries (2,000 resamples). better / worse = the 95% CI of the MRR@10 change excludes 0.

strategy	Δ MRR@10	95% CI	verdict
17_ltr_cv	0.279	[+0.245, +0.312]	better
13_router_bge	0.270	[+0.236, +0.305]	better
15_d2q_router_quality	0.267	[+0.232, +0.303]	better
16_plan_router_quality	0.264	[+0.230, +0.298]	better
20_ai_router_bge	0.262	[+0.228, +0.298]	better
17_router_listwise	0.259	[+0.223, +0.295]	better
14_qwen3_router_quality	0.254	[+0.218, +0.290]	better
10_enriched_router_quality	0.251	[+0.215, +0.286]	better
13_router_qwen3	0.241	[+0.203, +0.277]	better
20_ai_router_quality	0.236	[+0.201, +0.271]	better
09_router_quality	0.236	[+0.201, +0.269]	better
10_enriched_router_fast	0.224	[+0.188, +0.259]	better
16_plan_router_fast	0.224	[+0.189, +0.256]	better
07_dao_ai_instructed_llm_rerank	0.209	[+0.167, +0.248]	better
09_router_fast	0.200	[+0.169, +0.230]	better
20_ai_router_fast	0.192	[+0.160, +0.223]	better
09_hybrid_id_routing	0.167	[+0.137, +0.197]	better
18_scale_router_bge_50	0.122	[+0.075, +0.167]	better
13_hybrid50_bge	0.089	[+0.068, +0.111]	better
07_dao_ai_instructed_flashrank	0.077	[+0.054, +0.100]	better
06_filter_ai_decide	0.073	[+0.032, +0.116]	better
10_enriched_rerank	0.066	[+0.032, +0.101]	better
10_enriched_full_text	0.066	[+0.046, +0.085]	better
06_ai_decide_noul	0.061	[+0.042, +0.079]	better
09_ai_decide_fast	0.053	[+0.037, +0.069]	better

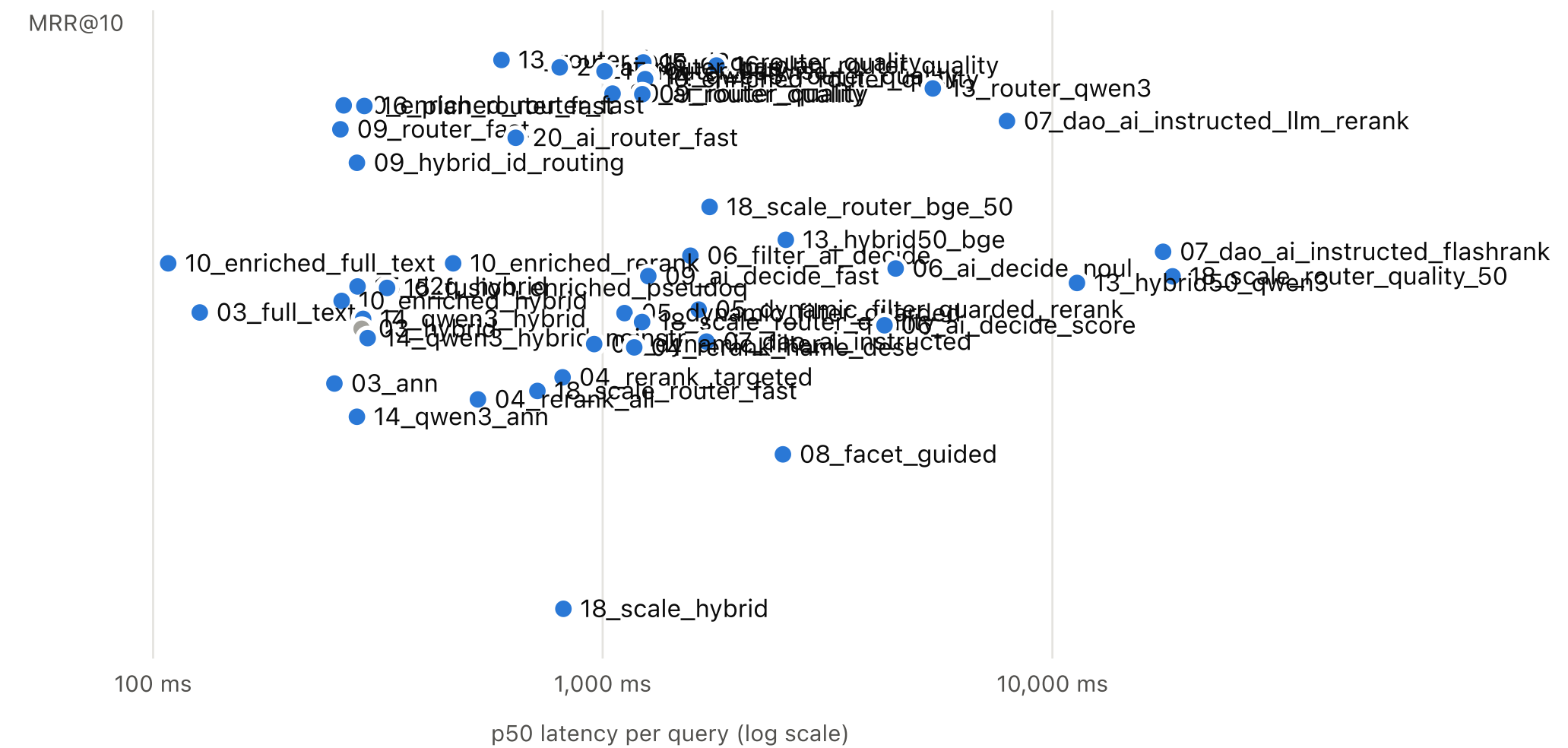
strategy	$\Delta\text{MRR@10}$	95% CI	verdict
18_scale_router_quality_50	0.053	[+0.004, +0.102]	better
13_hybrid50_qwen3	0.046	[+0.021, +0.072]	better
15_d2q_hybrid	0.042	[+0.024, +0.061]	better
15_fusion_enriched_pseudoq	0.041	[+0.025, +0.058]	better
10_enriched_hybrid	0.028	[+0.009, +0.046]	better
05_dynamic_filter_guarded_rerank	0.019	[-0.020, +0.058]	no clear difference
03_full_text	0.016	[-0.001, +0.034]	no clear difference
05_dynamic_filter_guarded	0.016	[-0.015, +0.046]	no clear difference
14_qwen3_hybrid	0.010	[-0.014, +0.034]	no clear difference
18_scale_router_quality	0.007	[-0.044, +0.057]	no clear difference
06_ai_decide_score	0.004	[-0.022, +0.028]	no clear difference
14_qwen3_hybrid_noinstr	-0.009	[-0.035, +0.016]	no clear difference
07_dao_ai_instructed	-0.013	[-0.041, +0.019]	no clear difference
05_dynamic_filter	-0.015	[-0.048, +0.017]	no clear difference
04_rerank_name_desc	-0.019	[-0.046, +0.008]	no clear difference
04_rerank_targeted	-0.049	[-0.078, -0.018]	worse
03_ann	-0.055	[-0.072, -0.037]	worse
18_scale_router_fast	-0.062	[-0.114, -0.012]	worse
04_rerank_all	-0.071	[-0.100, -0.044]	worse
14_qwen3_ann	-0.088	[-0.116, -0.060]	worse
08_facet_guided	-0.126	[-0.157, -0.096]	worse
18_scale_hybrid	-0.281	[-0.322, -0.241]	worse

Different query shapes, different winners

strategy (MRR@10 by query type)	known_item	descriptive	brand_constrained	attribute_constrained	exclusion	identifier
17_ltr_cv	0.970	0.890	0.911	0.963	0.902	1.000
13_router_bge	0.959	0.902	0.895	0.927	0.900	1.000
15_d2q_router_quality	0.973	0.871	0.902	0.932	0.890	1.000
16_plan_router_quality	0.963	0.872	0.910	0.943	0.861	1.000
20_ai_router_bge	0.959	0.902	0.875	0.927	0.875	1.000
17_router_listwise	0.954	0.839	0.904	0.934	0.883	1.000
14_qwen3_router_quality	0.967	0.843	0.902	0.939	0.834	1.000
10_enriched_router_quality	0.959	0.828	0.895	0.932	0.854	1.000
13_router_qwen3	0.895	0.874	0.855	0.920	0.868	1.000
20_ai_router_quality	0.925	0.820	0.882	0.915	0.838	1.000
09_router_quality	0.952	0.832	0.866	0.902	0.826	1.000
10_enriched_router_fast	0.918	0.752	0.881	0.924	0.834	1.000
16_plan_router_fast	0.929	0.789	0.864	0.922	0.801	1.000
07_dao_ai_instructed_llm_rerank	0.891	0.824	0.823	0.878	0.836	0.963
09_router_fast	0.845	0.808	0.847	0.874	0.792	1.000
20_ai_router_fast	0.845	0.808	0.827	0.874	0.762	1.000
09_hybrid_id_routing	0.823	0.812	0.832	0.861	0.635	1.000
18_scale_router_bge_50	0.882	0.551	0.716	0.869	0.680	1.000
13_hybrid50_bge	0.959	0.897	0.893	0.925	0.826	0.000
07_dao_ai_instructed_flashrank	0.966	0.848	0.873	0.912	0.827	0.003
06_filter_ai_decide	0.785	0.638	0.803	0.714	0.758	0.705
10_enriched_rerank	0.783	0.787	0.797	0.935	0.556	0.500
10_enriched_full_text	0.867	0.869	0.878	0.903	0.665	0.175
06_ai_decide_noul	0.908	0.866	0.871	0.922	0.760	0.000
09_ai_decide_fast	0.930	0.843	0.865	0.915	0.729	0.000
18_scale_router_quality_50	0.805	0.451	0.695	0.733	0.595	1.000

strategy (MRR@10 by query type)	known_item	descriptive	brand_constrained	attribute_constrained	exclusion	identifier
13_hybrid50_qwen3	0.890	0.877	0.854	0.919	0.700	0.000
15_d2q_hybrid	0.922	0.795	0.892	0.912	0.698	0.000
15_fusion_enriched_pseudoq	0.913	0.814	0.882	0.914	0.685	0.002
10_enriched_hybrid	0.898	0.750	0.877	0.908	0.667	0.032
05_dynamic_filter_guarded_rerank	0.656	0.740	0.770	0.762	0.799	0.352
03_full_text	0.849	0.837	0.852	0.887	0.637	0.000
05_dynamic_filter_guarded	0.735	0.630	0.829	0.758	0.755	0.351
14_qwen3_hybrid	0.907	0.766	0.889	0.886	0.472	0.102
18_scale_router_quality	0.714	0.405	0.672	0.677	0.536	1.000
06_ai_decide_score	0.885	0.769	0.818	0.874	0.639	0.000
03_hybrid	0.823	0.812	0.832	0.861	0.635	0.000
14_qwen3_hybrid_noinstr	0.872	0.718	0.874	0.875	0.478	0.090
07_dao_ai_instructed	0.705	0.770	0.728	0.844	0.796	0.044
05_dynamic_filter	0.702	0.651	0.723	0.692	0.750	0.354
04_rerank_name_desc	0.765	0.796	0.749	0.879	0.661	0.000
04_rerank_targeted	0.619	0.870	0.637	0.890	0.656	0.000
03_ann	0.799	0.700	0.800	0.764	0.571	0.000
18_scale_router_fast	0.666	0.324	0.621	0.552	0.425	1.000
04_rerank_all	0.647	0.816	0.657	0.879	0.538	0.001
14_qwen3_ann	0.885	0.556	0.865	0.788	0.340	0.000
08_facet_guided	0.753	0.572	0.763	0.574	0.547	0.000
18_scale_hybrid	0.666	0.294	0.561	0.503	0.056	0.198

Quality vs latency



Pick the cheapest strategy that is good enough per query shape; route only hard queries to the heavier pipeline.

Answer quality (end-to-end RAG)

gpt-oss-120b answers from each strategy's top-5; MLflow judges on the 100-query judged dataset.

strategy (judged subset)	correctness	retrieval_relevance	retrieval_sufficiency	retrieval_groundedness	answer_policy	mrr_at_10
03_hybrid	0.716	0.333	0.333	0.721		0.705
04_rerank_targeted	0.725	0.339	0.353	0.730		0.678
05_dynamic_filter_guarded_rerank	0.775	0.374	0.351	0.771		0.698
06_filter_ai_decide	0.804	0.335	0.338	0.716		0.753
07_dao_ai_instructed	0.775	0.117	0.716	0.895		0.698
09_router_quality	0.882	0.441	0.397	0.770		0.918

Holdout check

300 queries generated from products never used to build or tune the strategies. Rules, prompts and the learned ranker were developed on dev; these are the unbiased numbers.

strategy	dev MRR@10	holdout MRR@10	dev hit@1	holdout hit@1
17_ltr		0.934		0.890
20_ai_router_bge	0.923	0.920	0.885	0.880
17_router_listwise	0.919	0.920	0.878	0.873
13_router_bge	0.930	0.919	0.893	0.880
10_enriched_router_quality	0.911	0.909	0.865	0.857
09_router_quality	0.896	0.900	0.852	0.850
20_ai_router_quality	0.897	0.894	0.845	0.847
13_router_qwen3	0.902	0.893	0.845	0.827
09_router_fast	0.861	0.845	0.788	0.777
20_ai_router_fast	0.852	0.843	0.778	0.773
06_filter_ai_decide	0.734	0.740	0.652	0.670
03_full_text	0.677	0.690	0.607	0.627
03_hybrid	0.661	0.645	0.583	0.567

ai_decide as the router

Same retrieval and reranker in each pair; only the classifier differs. Rules = SKU/UPC regex + negation regex over the brand vocabulary. ai_decide = one call with two choice questions (route; excluded brand chosen from the query's top-10 brand facets). The paraphrase set rewrites the dev exclusion queries so the rules miss them ("I'm done with DeWalt").

strategy	classifier	dev MRR@10	holdout MRR@10	paraphrase MRR@10	paraphrase brand leak	dev p50_ms
09_router_fast	rules	0.861	0.845	0.591	81.0%	261
20_ai_router_fast	ai_decide	0.852	0.843	0.737	13.0%	640
09_router_quality	rules	0.896	0.900	0.691	74.0%	1,225
20_ai_router_quality	ai_decide	0.897	0.894	0.790	14.0%	1,051
13_router_bge	rules	0.930	0.919	0.771	83.0%	596
20_ai_router_bge	ai_decide	0.923	0.920	0.855	14.0%	802

How much of the remaining error is real?

Ground truth is the seed product a query was generated from. An ESCI-style LLM judge (Claude Sonnet 5.5) re-labels rank-1 results that aren't the seed: judged exact@1 counts the seed or a product judged an exact match. LLM judges are biased; treat as an estimate.

strategy	seed hit@1		judged exact@1		graded nDCG@5
03_hybrid	0.583		0.636		0.765
09_router_fast	0.788		0.850		0.767
09_router_quality	0.852		0.912		0.777
10_enriched_router_quality	0.865		0.923		0.773

Scale: 38K → ~5M products (storage-optimized endpoint)

Realistic distractors (brand swapped within category, sizes/counts perturbed) added to the real catalog; same strategies, only the endpoint type changes. The retailer's production index (~100M) is ~20x larger again. Deeper candidate pools (50) and a fast cross-encoder recover much of the loss.

strategy at ~5M	38K counterpart	MRR@10 38K	MRR@10 ~5M	hit@10 ~5M	p50 ms ~5M	p99 ms ~5M
18_scale_hybrid	10_enriched_hybrid	0.689	0.380	0.537	818	4,179
18_scale_router_fast	10_enriched_router_fast	0.885	0.598	0.725	716	2,200
18_scale_router_quality	10_enriched_router_quality	0.911	0.667	0.735	1,223	4,331
18_scale_router_quality_50	10_enriched_router_quality	0.911	0.713	0.797	18,510	142,269
18_scale_router_bge_50	13_router_bge	0.930	0.783	0.847	1,729	3,491

Side by side: baseline vs improved

IMPROVEMENT · KNOWN_ITEM

“Stihl MS194T 14" gas chainsaw”

target: STIHL MS 194 T 14 in. 31.72 cc Gas Chainsaw (SKU 29541355)

03_hybrid MRR 0.00

- 1. **STIHL** STIHL MS 261 16 in. 50.2 cc Gas Chainsaw SKU 45166686
- 2. **STIHL** STIHL MS 250 18 in. Gas Chainsaw SKU 25900545
- 3. **STIHL** STIHL MS 162 16 in. Light 01 Bar Gas Chainsaw Picco Micro Mini 3 PMM3 3/8 in. SKU 44898991
- 4. **STIHL** STIHL MS 261 C-M 18 in. 50.2 cc Gas Chainsaw SKU 55914703
- 5. **STIHL** STIHL MS 251 18 in. 45.6 cc Gas Chainsaw SKU 61739314

15_d2q_router_quality MRR 1.00

- 1. **STIHL** STIHL MS 194 T 14 in. 31.72 cc Gas Chainsaw SKU 29541355 ✓ **expected** rerank score 1.00
- 2. **STIHL** STIHL MS 261 16 in. 50.2 cc Gas Chainsaw SKU 45166686 rerank score 0.00
- 3. **STIHL** STIHL MS 180 C-BE 16 in. 31.8 cc Gas Chainsaw SKU 86948524 rerank score 0.00
- 4. **STIHL** STIHL MS 251 18 in. 45.6 cc Gas Chainsaw SKU 61739314 rerank score 0.00
- 5. **STIHL** STIHL HT 56 C-E 10 in. 27.2 cc Gas Chainsaw/Pole Saw Combo SKU 03096307 rerank score 0.00

IMPROVEMENT · DESCRIPTIVE

“1 in. spade drill bit hex shank clean wood holes 16 in. length”

target: Diablo 1 in. X 16 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk (SKU 71350660)

03_hybrid MRR 0.00

- 1. **DIABLO** Diablo 1/2 in. X 16 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk SKU 78904521
- 2. **DIABLO** Diablo 9/16 in. X 6 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk SKU 71640365
- 3. **DIABLO** Diablo 5/8 in. X 16 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk SKU 86231350
- 4. **DIABLO** Diablo 5/16 in. X 6 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk SKU 38201417
- 5. **DIABLO** Diablo 7/8 in. X 6 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk SKU 16748477

13_router_bge MRR 1.00

- 1. **DIABLO** Diablo 1 in. X 16 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk SKU 71350660 ✓ **expected** rerank score 1.00
- 2. **DIABLO** Diablo 3/4 in. X 16 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk SKU 72319895 rerank score 0.99
- 3. **DIABLO** Diablo 1/2 in. X 16 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk SKU 78904521 rerank score 0.99
- 4. **DIABLO** Diablo 5/8 in. X 16 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk SKU 86231350 rerank score 0.99
- 5. **DIABLO** Diablo 3/8 in. X 16 in. L High Speed Steel Clean Wood Spade Bit Hex Shank 1 pk SKU 76831607 rerank score 0.99

IMPROVEMENT · BRAND_CONSTRAINED

“Home Plus hand pump sprayer 48 oz with adjustable nozzle”

target: Home Plus 48 oz Sprayer Pump Hand Held Pump Sprayer (SKU 95360492) · constraints {"brand_name": "HOME PLUS"}

03_hybrid MRR 0.00

- 1. **HOME PLUS** Home Plus 2 Pattern Adjustable Plastic Hose Nozzle SKU 55265034
- 2. **HOME PLUS** Home Plus 7 Pattern Adjustable Multi-Pattern Plastic Hose Nozzle SKU 18735710
- 3. **STORE BRAND** Store Brand 2 gal Pump Deck & Bleach Sprayer SKU 39948809 ✗ **violates request**
- 4. **STORE BRAND** Store Brand 2 gal Sprayer Pump Lawn and Garden Sprayer SKU 46554123 ✗ **violates request**
- 5. **STORE BRAND** Store Brand 1 gal Sprayer Pump Lawn and Garden Sprayer SKU 30948549 ✗ **violates request**

17_ltr_cv MRR 1.00

- 1. **HOME PLUS** Home Plus 48 oz Sprayer Pump Hand Held Pump Sprayer SKU 95360492 ✓ **expected**
- 2. **HOME PLUS** Home Plus 48 oz Sprayer Pump Hand Held Pump Sprayer SKU 06631895 ✓ **expected**
- 3. **ORTHO** Ortho 48 oz Hand Held Spray Bottle SKU 24269469 ✗ **violates request**
- 4. **HOME PLUS** Home Plus 2 Pattern Adjustable Plastic Hose Nozzle SKU 55265034
- 5. **HOME PLUS** Home Plus 7 Pattern Adjustable Multi-Pattern Plastic Hose Nozzle SKU 18735710

IMPROVEMENT · ATTRIBUTE_CONSTRAINED

“handheld showerhead 1.6gpm 4-setting chrome 1/2-in connection”

target: Waterpik EcoFlow Chrome 4 settings Handheld Showerhead 1.6 gpm (SKU 51976125) · constraints {"merchandise_class": "SHOWER HEADS"}

03_hybrid MRR 0.00

- 1. **MOEN** Moen Engage Chrome 6 settings Handheld Showerhead 1.75 gpm SKU 74128433
- 2. **OAKBROOK** OakBrook Chrome PVC 5 settings Handheld Showerhead 1.8 gpm SKU 00500722
- 3. **MOEN** Moen Engage Chrome 6 settings Handheld Showerhead 1.75 gpm SKU 75702472
- 4. **WATERPIK** Waterpik PowerPulse Chrome Six settings Handheld Showerhead 1.8 gpm SKU 62614955
- 5. **CULLIGAN** Culligan Chrome Chrome 5 settings Handheld Showerhead 1.8 gpm SKU 85289567

17_ltr_cv MRR 1.00

- 1. **WATERPIK** Waterpik EcoFlow Chrome 4 settings Handheld Showerhead 1.6 gpm SKU 51976125 ✓ **expected**
- 2. **WATERPIK** Waterpik EcoFlow Chrome 4 settings Showerhead 1.6 gpm SKU 39861672
- 3. **MOEN** Moen Engage Chrome 6 settings Handheld Showerhead 1.75 gpm SKU 74128433
- 4. **MOEN** Moen Engage Chrome 6 settings Handheld Showerhead 1.75 gpm SKU 75702472
- 5. **OAKBROOK** OakBrook Chrome PVC 3 settings Handheld Showerhead 1.8 gpm SKU 16174917

IMPROVEMENT · EXCLUSION

“high strength clear glue 16 oz not GORILLA”

target: Elmer's Glue-All High Strength Polyvinyl acetate homopolymer Clear All Purpose Adhesive 16 oz (SKU 78510562) · constraints {"merchandise_class": "GLUES/ADHESIVES/APPLICATR", "exclude_brand": "GORILLA"}

03_hybrid MRR 0.00

- 1. **GORILLA** Gorilla High Strength Clear Clear Glue 1.75 oz SKU 62204491 ✖ **violates request**
- 2. **GORILLA** Gorilla High Strength Clear Clear Glue 3.75 oz SKU 75540111 ✖ **violates request**
- 3. **GORILLA** Gorilla High Strength Glue Clear Super Glue 0.22 oz SKU 16639843 ✖ **violates request**
- 4. **GORILLA** Gorilla High Strength Glue Clear Super Glue 0.11 oz SKU 18178098 ✖ **violates request**
- 5. **GORILLA** Gorilla High Strength Clear All Purpose Adhesive 5.75 oz SKU 09679498 ✖ **violates request**

17_itr_cv MRR 1.00

- 1. **ELMER'S** Elmer's Glue-All High Strength Polyvinyl acetate homopolymer Clear All Purpose Adhesive 16 oz SKU 78510562 ✔ **expected**
- 2. **ELMER'S** Elmer's Glue-All High Strength Polyvinyl acetate homopolymer Clear All Purpose Adhesive 7.63 oz SKU 97475309
- 3. **ELMER'S** Elmer's Glue-All High Strength Polyvinyl acetate homopolymer Clear All Purpose Adhesive 1 gal SKU 29340002
- 4. **LOCTITE** Loctite Extreme High Strength Crystal Clear Glue 1.62 oz SKU 42284307
- 5. **LOCTITE** Loctite Professional High Strength Glue Clear Super Glue 0.71 oz SKU 73531807

filters: [{"column": "brand_name", "op": "ne", "value": "GORILLA"}]

IMPROVEMENT · IDENTIFIER

“Do you have item 00222419?”

target: Scotts Turf Builder Tall Fescue Grass Sun or Shade Grass Seed 7 lb (SKU 00222419)

03_hybrid MRR 0.00

- 1. **LEVITON** Leviton Decora 15 amps 125 V Duplex Ivory Outlet 5-15R 1 pk SKU 49077254
- 2. "3"" Ceramic Angel in 12 Pi" SKU 39810632
- 3. **LEVITON** Leviton 15 amps 125 V Duplex Ivory Outlet 5-15R 10 pk SKU 53423757
- 4. **LEVITON** Leviton 15 amps 125 V Duplex Ivory Tamper Resistant Outlet 5-15R 1 pk SKU 18709681
- 5. **LEVITON** Leviton Decora 20 amps 125 V Duplex Ivory Outlet 5-20R 1 pk SKU 16008663

09_hybrid_id_routing MRR 1.00

- 1. **SCOTTS** Scotts Turf Builder Tall Fescue Grass Sun or Shade Grass Seed 7 lb SKU 00222419 ✔ **expected**

REGRESSION (HONEST VIEW) · DESCRIPTIVE

"bourbon flavored steak seasoning for grilling and burgers"

target: Weber Sazerac Buffalo Trace Seasoning 6.5 oz (SKU 58474815)

03_hybrid MRR 1.00

1. **WEBER** Weber Sazerac Buffalo Trace Seasoning 6.5 oz SKU 58474815 ✓ **expected**
2. **BIG** Bull Shit Steak Seasoning SKU 56759402
3. **BORSARI** Borsari Savory Seasoning Salt 4 oz SKU 07495502
4. **BORSARI** Borsari Cracked Pepper Seasoning Salt 3 oz SKU 29536702
5. **BLACKSTONE** Blackstone Steak & Burger Seasoning 8.7 oz SKU 38815959

17_itr_cv MRR 0.33

1. **BIG** Bull Shit Steak Seasoning SKU 56759402
2. **BORSARI** Borsari Cracked Pepper Seasoning Salt 3 oz SKU 29536702
3. **WEBER** Weber Sazerac Buffalo Trace Seasoning 6.5 oz SKU 58474815 ✓ **expected**
4. **BLACKSTONE** Blackstone Steak & Burger Seasoning 8.7 oz SKU 38815959
5. **BORSARI** Borsari Savory Seasoning Salt 4 oz SKU 07495502

REGRESSION (HONEST VIEW) · BRAND_CONSTRAINED

"Can Gun spray gun 1psi recycled plastic"

target: Can Gun 1 1 psi Recycled Plastic Spray Gun (SKU 40966205) · constraints {"brand_name": "CAN GUN"}

03_hybrid MRR 1.00

1. **CAN GUN** Can Gun 1 1 psi Recycled Plastic Spray Gun SKU 40966205 ✓ **expected**
2. **CAN GUN** Can Gun 1 psi Recycled Plastic Airless Spray Gun SKU 42215646
3. **SURFACEMAXX** SurfaceMaxx Pressure Washer Gun 4500 psi SKU 04755661 ✗ **violates request**
4. **WAGNER SPRAY TECH** Wagner Power Painter Plus 2200 psi Plastic Airless Paint Sprayer SKU 08552273 ✗ **violates request**
5. **WAGNER** Wagner Furno 500 12.5 amps 1500 W 120 V Digital Heat Gun SKU 67939497 ✗ **violates request**

17_itr_cv MRR 0.50

1. **CAN GUN** Can Gun 1 psi Recycled Plastic Airless Spray Gun SKU 42215646
2. **CAN GUN** Can Gun 1 1 psi Recycled Plastic Spray Gun SKU 40966205 ✓ **expected**
3. **WAGNER SPRAY TECH** Wagner Power Painter Plus 2200 psi Plastic Airless Paint Sprayer SKU 08552273 ✗ **violates request**
4. **WAGNER SPRAY TECH** Wagner Power Painter 1800 psi Plastic Airless Paint Sprayer SKU 70250404 ✗ **violates request**
5. **SURFACEMAXX** SurfaceMaxx 6-in-1 Change-Over Spray Nozzle 3600 psi SKU 40305714 ✗ **violates request**

Recommendations

1. **Route by query shape first; it's the biggest, cheapest win.** SKU/UPC → exact filter, "not X" → `brand_name NOT` filter parsed from the text (no LLM), else cleaned HYBRID. Fast router: MRR 0.845 on holdout at ~261 ms p50 vs HYBRID 0.645.
2. **Rerank the router's candidates with a GPU cross-encoder.** Router + bge-reranker-v2-m3 (Model Serving, 25 candidates): MRR 0.930 dev / 0.919 holdout at ~596 ms p50, versus the ai_decide quality router 0.896 / 0.900 at ~1.2s. Qwen3-Reranker-0.6B was slower and no better on a small GPU.
3. **For the last points, learn the ranking.** A LightGBM LambdaRank model over router candidates (pool/ANN/FULL_TEXT ranks, ai_decide, two cross-encoders, lexical features) reached MRR 0.939 (5-fold CV on dev) and 0.934 on holdout, the best result, but at ~3.6s p50 because every signal is computed per request; distil to the few strongest signals for production.
4. **Embed and enrich what shoppers search for.** A compact title | brand | category | attributes | codes field helped every strategy (router 0.896 → 0.911); Doc2Query pseudo-queries added a little more (0.928) but are likely optimistic on this LLM-generated eval set. Qwen3 instruction-aware embeddings matched gte-large under the router (0.914).
5. **Keep LLMs out of the hot path unless they earn it.** An LLM structured query plan scored 0.925 but leaked the excluded brand on 12% of exclusion queries vs 1% for the rules; LLM-planned filters varied run to run. Use them for the long tail the rules don't cover, with catalog validation and caching.
6. **Keep the rules as the router; add ai_decide only as a fallback.** Classifying the route with one ai_decide call (`create_router_tool(..., classifier="ai_decide")`) tied the rules on holdout (0.920 vs 0.919 with bge) but missed some plain "not X" exclusions (brand leak 10% vs 1%) and added ~206 ms p50. On paraphrased exclusions the rules can't parse ("I'm done with DeWalt") it lifted MRR nan → nan. Run it only when the rules find no exclusion but the query names a catalog brand.
7. **Plan for scale: depth matters.** On a ~5M-row storage-optimized index with near-duplicate distractors, the 12-candidate quality router fell to MRR 0.667 (hit@10 74%); 50 candidates: 0.713 with ai_decide, 0.783 with the bge cross-encoder. At 100M, budget deeper candidate pools and a fast cross-encoder.
8. **Measure honestly.** An ESCI-style LLM judge found ~6 points of apparent rank-1 misses were valid alternatives (judged exact@1 92% for the enriched quality router), so true rank-1 accuracy is higher than hit@1 shows; confirm on real production query logs with graded labels before rollout. The MLflow dataset, scorers and traces make every change a measured experiment.

Reproduce & explore in MLflow

- Experiment: `/Users/<user>/ai-search-product-retrieval/retrieval_experiments`. Open **Evaluations**, select two runs, then **Compare** for row-level side by side.
- Every run evaluates the same MLflow dataset `retail_consumer_goods.product_search.retrieval_eval`, so rows align across runs.
- Notebooks: `/Users/<user>/ai-search-product-retrieval` (00–91). Library versions: {"product-retrieval": "0.1.0", "databricks-langchain": "0.20.0", "databricks-ai-search": "0.78", "databricks-sdk": "0.150.0", "mlflow": "3.17.0"}

run	run id
03_ann	c48d14a0bdaf4044bcebf9d607c96405
03_full_text	755ddba75e9c4364b20e54be2cc42387
03_hybrid	00745cd707cb4d319872bebcd5516aff
04_rerank_all	23f2bb92596e44e4883cee6282cdc771
04_rerank_name_desc	7ccb0163f1354c6987c7b6aa32a70f27
04_rerank_targeted	a032263aec9a4255ae3623709632371a
05_dynamic_filter	eaf02a92d2b14f828600badcbb03e9e7
05_dynamic_filter_guarded	c156dfe2996c46288837f158912a7c22
05_dynamic_filter_guarded_rerank	cc7a114ed91540609e02228c78b8e45a
06_ai_decide_noul	de45a2643ad84aaca34fbee1a70d7966
06_ai_decide_score	01b860ee62ef4ba391684f9efc9cd565
06_filter_ai_decide	a6b70ca9142b497eba8affb458ead251
07_dao_ai_instructed	25a8c7f8d0aa4143a3094462c8af2f2c
07_dao_ai_instructed_flashrank	a7e8a2ab0e1c4e02843fc81b9c75f713
07_dao_ai_instructed_llm_rerank	b527e3c70cfc402e912a23f3a7cd8443
08_facet_guided	5a043c4c7bcd4e488c4244e06fad7e9c
09_ai_decide_fast	91e91304b872472c9c1716f76b711a4b
09_hybrid_id_routing	062f2cafcf3047229ba55cd11ab4c33a
09_router_fast	7428b0a3fde44c6395e2f4b8d9d92768
09_router_quality	2787ba5f21c34b7687c0d8c48adcc5de

10_enriched_full_text	e4c03a6e88474859b1cf34402bd52f20
10_enriched_hybrid	2f5c0c72533f4690aacfb5106a24793c
10_enriched_rerank	1586e0360bf645598553f8c027387452
10_enriched_router_fast	89847ed5e1ff40d5871324b874b181be
10_enriched_router_quality	21498e9b30d242398c6cf93aa96982f3
13_hybrid50_bge	b4469eb63f754a2eb39e96d672ecd4cd
13_hybrid50_qwen3	0c54251a5c2f463db89d2e86bfba1322
13_router_bge	641dc5b505fd4aad894b2db23d8bf1be
13_router_qwen3	11e25cfb44a3495e81d135ea16c6d4d5
14_qwen3_ann	198b00431a94446792d6fd320310bf3c
14_qwen3_hybrid	410ff13f077f4cf5978410d03ff1d894
14_qwen3_hybrid_noinstr	5584d392120142e4910dc04cebf97127
14_qwen3_router_quality	d9f7b31aca3b494da9de28c2351542f2
15_d2q_hybrid	86c9cf6ffeeb4339b706cea5039dd8d3
15_d2q_router_quality	2f0895b6313c4aaeb0b795dfadeb35f5
15_fusion_enriched_pseudoq	38d20524a73d4d6da5c253aa63bc4a41
16_plan_router_fast	24b2cd17f78b49139712b27292a0f582
16_plan_router_quality	a26a85cf77b8418caf45d9f2d97873e3
17_ltr_cv	c99490684f2f44be8c4dde4a6c73c9cd
17_router_listwise	5c916a6f10094f7799a6b82b940ea87d
18_scale_hybrid	8ab50a032b3746bda3317e5c32ce7d7f
18_scale_router_bge_50	bab3c782b72b4b92a4a7d4e9367e8d6d
18_scale_router_fast	bbd28d49f7024fc3bbd3387c1825ed79
18_scale_router_quality	92a2c1447cd046dd8e0bac122c831a42
18_scale_router_quality_50	10721246742a414eb3ed0d73228c800b
20_ai_router_bge	695dda3ec7b84ae2873ab286a7c1f870
20_ai_router_fast	e6983f04d6c24f96adc05fde7543ce
20_ai_router_quality	f181011c90a2441eb699e41ad97bafb3